TASK 1.1 REPORT

Market Data Analysis and Visualisation

1. Data Methodology

SECTOR	STOCK	TICKER
Banking Sector (BANK)	HDFC Bank Limited	HDFCBANK.NS
	ICICI Bank Limited	ICICIBANK.NS
	Axis Bank Limited	AXISBANK.NS
Information Technology Sector (IT)	Tata Consultancy Services Limited	TCS.NS
	Infosys Limited	INFY.NS
	Wipro Limited	WIPRO.NS
Energy Sector (ENERGY)	Reliance Industries Limited	RELIANCE.NS
	Oil and Natural Gas Corporation Limited	ONGC.NS
	Bharat Petroleum Corporation Limited	BPCL.NS
Automotive Sector (AUTO)	Maruti Suzuki India Limited	MARUTI.NS
	Hero MotoCorp Limited	HEROMOTOCO.NS
	Mahindra & Mahindra Limited	M&M.NS

- **Date Range :** April 1, 2021 to March 31, 2026 (Same for All)
- **Source :** Yahoo Finance
- **Cleaning Steps :**
    1. Filled missing Adjusted Closed prices with the preceding value via ffill.
    2. Any null entry detected within the daily Volume column was filled by 0.
    3. Eliminated duplicate date entries.

4. All twelve cleaned dataframes and the NIFTY 50 benchmark dataframe were merged using a multi-lateral **Inner Join** matching strictly on the `Date` index.

2. Dashboard Section

1. **AXISBANK.NS (Banking Sector) :**
    - **Volatility Spikes:** critical risk regime shifts with multiple high-risk deviations exceeding +2σ above the mean, clustering heavily early 2022 and late 2022 where 20-day rolling volatility breached.
    - **Visible Trend:** The asset exhibits a structural upward trend punctuated by an extended consolidation phase throughout 2021, a strong cyclical bull run across 2022–2024, and a late-stage mean-reversion correction starting in late 2025.
    - **Volume Confirmation:** Major price breakout points, specifically the trend acceleration in late 2022 and early 2024 show strong volume confirmation, with daily trading volumes spiking toward 4×10^7 to 6×10^7 shares traded.
2. **HDFCBANK.NS (Banking Sector) :**
    - **Outperformance vs. Benchmark:** HDFC Bank severely lagged the broader market, serving as a significant drag on portfolio performance. It failed to maintain pace with the NIFTY 50, ultimately mean-reverting back to its baseline entry index of 100 by the end of the 5-year period.
    - **Volatility Spikes:** Risk anomalies are highly localized, with a severe volatility breakout occurring in early 2022 and a secondary structural risk expansion manifesting during the steep price sell-off in early 2026.
    - **Visible Trend:** The asset shows an uncharacteristic lack of structural growth.
    - **Volume Confirmation:** Price directions are confirmed by volume, but negatively.
3. **ICICIBANK.NS (Banking Sector) :**
    - **Outperformance vs. Benchmark:** ICICI Bank was an elite compounding asset, delivering consistent, powerful outperformance over the benchmark throughout the entire five-year timeline.

- **Volatility Spikes:** High-risk standard deviation anomalies (>+2σ) were strictly constrained to the market-wide stress windows of late 2021 and mid-2022.
- **Visible Trend:** The asset boasts the most consistent, linear, and institutional-grade upward trend, characterized by shallow drawdowns and a reliable 20-day moving average acting as dynamic support.
- **Volume Confirmation:** Volumetric accumulation is evident across the entire timeline.

4. **BPCL.NS (Energy Sector) :**
    - **Outperformance vs. Benchmark:** BPCL dramatically outperformed the NIFTY 50 baseline, experiencing a massive structural breakout in late 2023 that propelled its cumulative value from an underperforming base to an outperforming asset.
    - **Volatility Spikes:** The asset remained highly stable until early 2024, where it suffered its most severe volatility expansion.
    - **Visible Trend:** The chart documents a clear regime change: a flat, multi-year bearish accumulation phase from 2021 to mid-2023, followed by an aggressive, near-vertical momentum trend through 2024, before entering a sharp distribution drawdown in early 2026.
    - **Volume Confirmation:** The monumental price expansion in 2024 is heavily validated by the volume subplot.
5. **ONGC.NS (Energy Sector) :**
    - **Outperformance vs. Benchmark:** ONGC generated massive structural alpha relative to the benchmark.
    - **Volatility Spikes:** As a cyclical commodity asset, it exhibited extreme risk metrics in both mid-2022 and mid-2024, representing some of the highest localized risk numbers in the dataset.
    - **Visible Trend:** A slow ascending baseline from 2021 to 2023, a massive structural vertical squeeze throughout 2024, and a highly volatile, high-level horizontal distribution box across 2025–2026.
    - **Volume Confirmation:** The massive price rerating in late 2023 and early 2024 is perfectly synchronized with an absolute expansion in trading volume.
6. **RELIANCE.NS (Energy Sector) :**
    - **Outperformance vs. Benchmark:** Reliance Industries mirrored the NIFTY 50 benchmark closely for the majority of the timeline.
    - **Volatility Spikes:** Volatility characteristics were tightly managed and non-erratic, with localized risk expansions peaking in late 2021

and mid-2022, followed by a long, flat period of low-volatility tracking.
- **Visible Trend:** The heavy-weight stock exemplifies a multi-step staircase trend.
- **Volume Confirmation:** Price breakouts out of the primary ranges are well-supported by volume expansions.

7. **M&M.S (Automotive Sector) :**
   - **Outperformance vs. Benchmark:** Mahindra & Mahindra emerged as the ultimate alpha generator of the entire 12-asset universe, completely outpacing the NIFTY 50.
   - **Volatility Spikes:** Despite the spectacular price velocity, volatility remained remarkably controlled.
   - **Visible Trend:** The chart outlines an impeccable, structurally flawless exponential growth curve.
   - **Volume Confirmation:** The structural advance is heavily backed by volume confirmation.
8. **MARUTI.NS (Automotive Sector) :**
   - **Outperformance vs. Benchmark:** Maruti Suzuki delivered robust alpha, steadily outpacing the NIFTY 50 index.
   - **Volatility Spikes:** The asset experienced severe volatility shocks during the first half of the timeline in early 2022 before entering a long-term structural volatility dampening regime.
   - **Visible Trend:** The long-term price action depicts a highly defined, multi-year ascending channel.
   - **Volume Confirmation:** Late 2025 trend exhaustion shows clear volume exhaustion.
9. **HEROMOTOCO.NS (Automotive Sector) :**
   - **Outperformance vs. Benchmark:** Hero MotoCorp demonstrated spectacular long-term alpha generation, handily beating the NIFTY 50.
   - **Volatility Spikes:** The primary volatility regime shock occurred in early 2022. Outside of this window, the stock maintained a stable, well-behaved risk profile.
   - **Visible Trend:** The baseline displays a rounded, multi-year cup-and-handle accumulation base spanning 2021 through mid-2023, which seamlessly transitioned into a powerful structural uptrend across 2024 and 2025.
10. **INFY.NS (Information Technology Sector) :**
    - **Outperformance vs. Benchmark:** Infosys underperformed the benchmark index over the comprehensive 5-year cycle.

- **Volatility Spikes:** Chart 4 displays persistent, erratic risk behavior with frequent, scattered volatility spikes. The most extreme risk expansion occurred in mid-2023.
- **Visible Trend:** The equity experienced a highly cyclical "boom-and-bust" trajectory: an initial bullish expansion in late 2021, a severe bear market drawdown throughout 2022–2023, a secondary recovery peak in 2025, and a steep macro collapse in early 2026.
- **Volume Confirmation:** Price direction is heavily confirmed by volume dynamics.

11. **TCS.NS (Information Technology Sector) :**
    - **Outperformance vs. Benchmark:** TCS significantly underperformed the NIFTY 50 benchmark over the comprehensive 5-year macro cycle.
    - **Volatility Spikes:** Chart 4 isolates localized risk regime shifts, with the most extreme volatility spikes cluster-forming in mid-2022 and early 2026.
    - **Visible Trend:** The asset experienced a bullish advance to an intermediate peak in 2021, a long cyclical bear market throughout 2022–2023, a secondary recovery in early 2025, and a final vertical collapse through early 2026.
    - **Volume Confirmation:** Structural trend changes are heavily confirmed by volume.
12. **WIPRO.NS (Information Technology Sector) :**
    - **Outperformance vs. Benchmark:** After an initial speculative surge in 2021, the stock spent the remaining four years of the cycle deeply trailing the benchmark, ultimately finishing the tracking window back at its nominal baseline entry index.
    - **Volatility Spikes:** The asset exhibited highly erratic risk characteristics.
    - **Visible Trend:** Multi-year "boom, bust, and stagnation" regime. Wipro staged an aggressive vertical rally in late 2021, followed by a protracted, 18-month bear market that erased all gains in 2023, before spending 2024-2026 in a volatile, lower-high horizontal range.
    - **Volume Confirmation:** Volumetric confirmation is visible at key inflection points.

SUMMARY : The **Automotive (AUTO)** sector was the ultimate winner, with every single one of its companies beating the benchmark by a wide margin. The

Energy (ENERGY) sector also did incredibly well, with companies like ONGC.NS and BPCL.NS.On the other hand, the **Information Technology (IT)** sector consistently lagged behind and was the worst performer. Tech companies suffered from a "boom-and-bust" cycle where they gained value early on but then lost it all.

3. Correlation Section

Analysis Questions Answer :

1. **Most Correlated Sector Pair** : ***AUTO*** and ***BANK*** (0.31). While a correlation of 0.31 is moderately low in absolute terms, still for capital allocation and sector rotation strategies, this relationship implies that transitioning capital directly between Auto and Banking will not provide high-quality structural diversification during macroeconomic shifts.
2. **Stock with lowest average correlation with the rest :** Stock : *Oil and Natural Gas Corporation Limited,* Average Correlation with Rest: *0.2095*
3. **Unexpected Cross-Sector Relations :** All cross-sector Correlations are less than 0.31 and greater than 0.20, indicating that there is no such unexpected relations where stock of one sector behaves like that of another sector.
4. If a client holds only banking stocks, IT sector can reduce their risk the most, because it has the lowest correlation (0.20) with the banking sector.

Pairs Based on Threshold :

1. **>0.80 =** 0 pairs
   Interpretation : This indicates that no two individual equities in this are moving in near-perfect lockstep. The portfolio is structurally clean and free of redundant asset allocations, meaning every chosen stock adds unique risk-return characteristics rather than merely duplicating another asset's movements.
2. **<0.20 =** 14 pairs

STOCK 1	**STOCK 2**	**CORRELATION**	**INTERPRETATION**
ENERGY_ONGC.NS	AUTO_HEROMOTOCO.NS	0.17	Strong commodity vs. domestic consumer

			discretionary hedge.
ENERGY_ONGC.NS	AUTO_M&M.NS	0.19	Decoupled industrial growth vs. global resource discovery vector.
ENERGY_ONGC.NS	AUTO_MARUTI.NS	0.14	Excellent hedge; high fuel prices hurt autos but help upstream energy producers.
ENERGY_ONGC.NS	BANK_HDFCBANK.NS	0.14	insulates portfolio from local banking system credit cycles.
ENERGY_ONGC.NS	IT_INFY.NS	0.13	Absolute lowest correlation; blends global tech spend with energy markets.
ENERGY_ONGC.NS	IT_TCS.NS	0.14	Combines defensive IT services with highly cyclical energy revenues.
IT_INFY.NS	AUTO_HEROMOTOCO.NS	0.17	Separates international US Dollar (USD) revenue from rural Indian credit cycles.
IT_INFY.NS	AUTO_MARUTI.NS	0.15	Excellent balance between urban consumer credit and global tech demand.
IT_INFY.NS	ENERGY_BPCL.NS	0.16	Pairs a margin-stable tech exporter with a domestic energy refiner.
IT_TCS.NS	AUTO_MARUTI.NS	0.17	Pairs heavy institutional blue-chip exporters with retail auto sales.

IT_TCS.NS	BANK_AXISBANK.NS	0.19	Effective offset for domestic corporate credit banking risk.
IT_WIPRO.NS	AUTO_MARUTI.NS	0.16	Combines consumer discretionary auto trends with global software demand.
BANK_ICICIBANK.NS	ENERGY_ONGC.NS	0.19	Symmetric verification of the Maruti-ONGC diversification channel
IT_INFY.NS	BANK_AXISBANK.NS	0.19	Provides a solid buffer against domestic corporate banking credit stress.

Interpretation : These 14 pairs provide institutional-grade diversification options. By intentionally building asset allocations around these specific pairs, portfolio managers can construct an equity portfolio that significantly reduces overall risk without sacrificing upside potential. This works because the underlying drivers of these stocks are largely independent of one another.

4. Volatility Section

- **Sector Ranking** (From most stable to least stable) :
    1. **Banking (BANK) - Rank 1 (Most Stable):** Annualized Volatility = **21.98%**
    2. **Information Technology (IT) - Rank 2:** Annualized Volatility = **23.70%**
    3. **Automotive (AUTO) - Rank 3:** Annualized Volatility = **25.55%**
    4. **Energy (ENERGY) - Rank 4 (Least Stable):** Annualized Volatility = **27.10%**
- **Stress Event Analysis :**

    1. **Volatility Multiplier Ratio :** This ratio compares a sector's short-term volatility during a crisis against its long-term average baseline. A ratio greater than 1.0 indicates that the crisis actively caused the sector's risk to expand beyond its normal levels.

SECTOR	RUSSIA-UKRAINE WAR	RBI RATE HIKE	ADANI CRISIS
AUTO	1.37	0.84	0.99
BANK	1.38	0.73	1.09
ENERGY	1.40	1.05	1.19
IT	1.15	1.10	1.07

Core Findings from the Ratios :

- **The Russia-Ukraine War:** This geopolitical shock acted as a universal risk amplifier across all industries. The Energy sector was hit the hardest, with its volatility exploding to **1.40 times** its typical baseline. This surge was triggered by immediate supply chain disruptions and global crude oil price shocks.
- **The RBI Rate Hike Shock:** This event created a clear split in the market. Domestic consumer-driven sectors like Auto (0.84) and Banking (0.73) actually experienced compressed or stable volatility. This shows that the interest rate hikes were already priced into the market. However, the IT sector's volatility expanded to **1.10 times** its normal level. This reflects the high sensitivity of tech stock valuations to rising global interest rates.
- **The Adani Short-Seller Crisis:** This localized market crisis primarily impacted the Energy (1.19) and Banking (1.09) sectors. These sectors saw an expansion in risk due to direct structural concerns over corporate leverage, overlapping infrastructure investments, and banking system loan exposures.

- **Quadrant Mapping Analysis : Risk-Return Ratio** (annualized return divided by annualized volatility) to find out which sector gives investors the most reward for every unit of risk they take on.

  WINNER : The **Automotive** sector secured the absolute optimal risk-return tradeoff verdict. It registered a premier return-per-unit-risk ratio of **0.8247**. Auto combined a very strong annualized return profile of

21.47% with a well-managed volatility baseline of 25.55%, proving to be an elite engine for portfolio growth.
RUNNER-UP : The **Energy** sector placed second with a solid risk-return ratio of **0.6851**. While it generated an impressive annualized return of 19.33%, its high volatility baseline of 27.10% held it back from beating the Auto sector on a risk-adjusted basis.
LAGGING DOMAINS :
Banking (0.5022): Settled into the Low Return / Low Volatility quadrant. It offered reliable downside protection but lacked the growth needed to generate strong outperformance over the 5-year period.
Information Technology (0.0656): Fell directly into the worst possible position—the Low Return / High Volatility quadrant. IT turned out to be a classic value trap, exposing investors to a high risk level while delivering a near-zero annualized return profile of just 1.71%.

- **Safest Sector :** The **Banking (BANK)** sector is the most safest sector over the five-year horizon, recording the lowest average annualized volatility at **21.98%**. This resilience was anchored by the steady, institutional compounding of `ICICIBANK.NS`. This stock helped insulate the broader sector from the larger volatility swings seen in the consumer discretionary and energy markets.
- **Best risk-return Tradeoff :** The **Automotive (AUTO)** sector offers the absolute best risk-return tradeoff, achieving a dominant return-per-unit-risk ratio of **0.8247**. This outperformance occurs because it generated the highest annualized return in the study at **21.47%** while keeping its annualized volatility restricted to **25.55%**. Unlike the Energy sector, which generated strong returns (19.33%) but exposed investors to very high risk (27.10%), the Auto sector expanded efficiently. This means it provided investors with maximum profitability for every unit of price uncertainty they took on over the five-year period.
- **Worst stress event with recovery speed :** The **Russia-Ukraine War** was the worst historical stress event in the dataset, triggering a sharp and universal risk expansion across all four sectors. During this crisis, the Volatility Multiplier Ratios rose significantly above their historical averages, led by **Energy at 1.40**, **Banking at 1.38**, and **Auto at 1.37**.

  The recovery speeds across the sectors showed a clear split based on their business models:

  1. **Fast Recovery (IT):** The Information Technology sector was the least disrupted, posting a low multiplier of **1.15**. Because IT

companies run on digital service contracts, their volatility normalized within a few weeks.
    2. **Slow Recovery (Energy, Auto, Banking):** These sectors suffered a prolonged risk expansion that took several months to stabilize. This delay was caused by direct exposure to physical disruptions, including global supply chain breakdowns, sudden spikes in raw material costs, and immediate liquidity interventions by central banks.
- **Stocks in the efficient quadrant :**
  An analysis of individual asset metrics reveals that **`ICICIBANK.NS` (Banking)** and **`MARUTI.NS` (Automotive)** are the clear standout stocks sitting in the high-efficiency zone.
    1. **`ICICIBANK.NS`** separated itself from its banking peers by combining low baseline volatility with a strong, upward price trend.
    2. **`MARUTI.NS`** maintained a well-behaved risk profile while delivering steady alpha to finish.
    3. While **`M&M.NS`** and **`ONGC.NS`** generated exceptional cumulative returns, their higher volatility baselines pushed them into the high-return/high-volatility opportunistic growth quadrant rather than the low-risk efficient core.
    4. Conversely, the IT stocks (`INFY.NS`, `TCS.NS`, `WIPRO.NS`) fell into the worst-performing low-return/high-volatility quadrant, showing they acted as clear value traps over this macro cycle.

5. Investment Conclusions

Portfolio Construction Recommendations :

1. **Overweight Allocation: Automotive (AUTO) :**
    - **Strategic Justification:** The Automotive sector represents the primary growth engine in the universe, commanding a top-tier Risk-Return Ratio of **0.8247** and a stellar annualized return profile of **21.47%**.
    - **Implementation Note:** Rather than equal-weighting, allocations should lean into structural leaders like `M&M.NS` (which displayed superior, exponential compounding up to an indexed value near 400) and `MARUTI.NS` (which serves as a highly efficient, lower-volatility anchor within the growth quadrant). This sector

provides robust consumer discretionary momentum that has fundamentally outpaced the broader NIFTY 50 benchmark.

2. **Overweight Allocation: Energy (ENERGY) :**
    - **Strategic Justification:** The Energy sector acts as an exceptional secondary alpha generator, achieving an annualized return of **19.33%** and a solid Risk-Return Ratio of **0.6851**.
    - **Implementation Note:** Because the sector presents the highest baseline annualized volatility at **27.10%** and is highly vulnerable to macro supply shocks (evidenced by its extreme 1.40 volatility multiplier during the Russia-Ukraine War), positions should be structurally paired with non-correlated counterparties. Specifically, combining upstream producers like `ONGC.NS` with highly decoupled partners like `IT_INFY.NS` (their pairwise correlation of **0.131** is the absolute lowest in the portfolio) allows the manager to harvest high commodity-driven returns while neutralizing tail-risk volatility.
3. **Neutral Allocation: Banking (BANK) :**
    - **Strategic Justification:** The Banking sector serves as the ultimate defensive ballast, recording the lowest baseline annualized volatility at **21.98%**. However, its return profile was significantly muted at **11.08%** due to severe internal performance dispersion.
    - **Implementation Note: Overweight** elite compounding assets like `ICICIBANK.NS`(which safely inhabited the high-return/low-volatility efficient quadrant), while structurally **Underweighting** or eliminating stagnating assets like `HDFCBANK.NS` which served as a dead-weight capital drag over the multi-year timeline.
4. **Underweight Allocation: Information Technology (IT) :**
    - **Strategic Justification:** The IT sector represents a classic value trap over this macroeconomic cycle, falling directly into the sub-optimal Low Return / High Volatility quadrant.
    - **Implementation Note:** All three constituent tech stocks (`INFY.NS`, `TCS.NS`, `WIPRO.NS`) suffered a destructive "boom-and-bust" cycle, failing to outpace the benchmark or preserve capital. IT allocations should be strictly limited to a minor tactical sleeve.

Framework For Extending the Analysis :

1. **Implementation of Multi-Factor Models (Fama-French) :** The current framework relies entirely on the capital asset pricing model (CAPM) and single-index relative returns. Extending this to a multi-factor environment will allow analysts to isolate the true drivers of alpha.

- **Methodology:** Regress the 12-asset return arrays against the Fama-French Indian Three-Factor or Five-Factor models. This process extracts explicit risk exposures to the Market Risk Premium, Size factor (Small Minus Big - SMB), Value factor (High Minus Low - HML), and Profitability/Investment vectors.
- **Value Add:** This maps out whether `M&M.NS`'s massive outperformance was driven by idiosyncratic corporate alpha, or if it was simply an unhedged systematic exposure to mid-cap value factors.

2. **Regime-Conditional Correlation Modelling :** Our current pairwise findings assume that correlations remain static over time. In real-world environments, asset correlations break down and shift dramatically during periods of high market stress.
    - **Methodology:** Deploy a Dynamic Conditional Correlation (DCC-GARCH) model or a Markov-Switching approach to isolate correlation matrices under distinct volatility regimes (e.g., "Normal Expansion" vs. "Liquidity Panic").
    - **Value Add:** This mathematical layer would explicitly quantify how cross-sector hedges alter during tail-risk events, preventing catastrophic portfolio drawdowns when asset correlations suddenly snap toward 1.0 during a global market liquidation.
3. **Forward-Looking Monte Carlo Asset Simulations :** The existing report looks exclusively backward at a single realized slice of history (2021–2026). To properly stress-test portfolio longevity for institutional clients, forward-looking predictive modeling is essential.
    - **Methodology:** Use the calculated historical return means, annualized volatilities, and the dense 12x12 Pearson correlation matrix to parametrize a multivariate Geometric Brownian Motion (GBM) simulation framework. Generate 10,000 randomized forward-looking price pathways over the next 3 to 5 years.
    - **Value Add:** This modeling compiles a dense probability distribution for final portfolio wealth. It allows advisors to visually demonstrate precise Value-at-Risk (VaR) thresholds, maximum drawdown expectations, and probability boundaries for achieving specific target returns, giving clients a clear, risk-tested roadmap for the future.